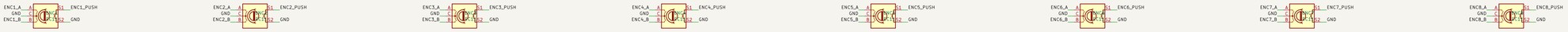
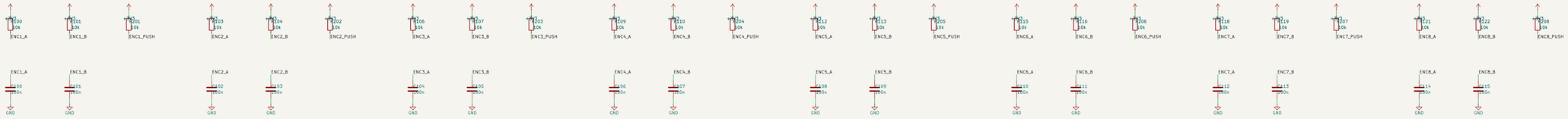
PANEL-1 Bukifon v1.0 (spec/20-electronics.md 6.1) 300 x 132 x 1.6 mm CERN-OHL-S-2.0 upper part Y 0..66, deck niche Y 66..145 (spec 10-panel-ui 3)

Coordinates of all controls = spec/10-panel-ui.md (ENC1..8, BTN_L1/L2/R1/R2, deck IDC-16, screen 1x14). Connectivity is net-label based.

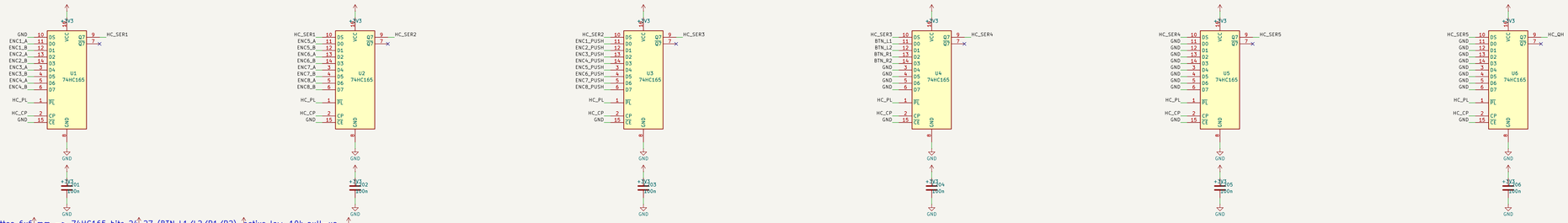


8 x EC11 (smooth, with push switch) + 16 x RC 10k/100n on A/B + 8 x 10k push pull-up





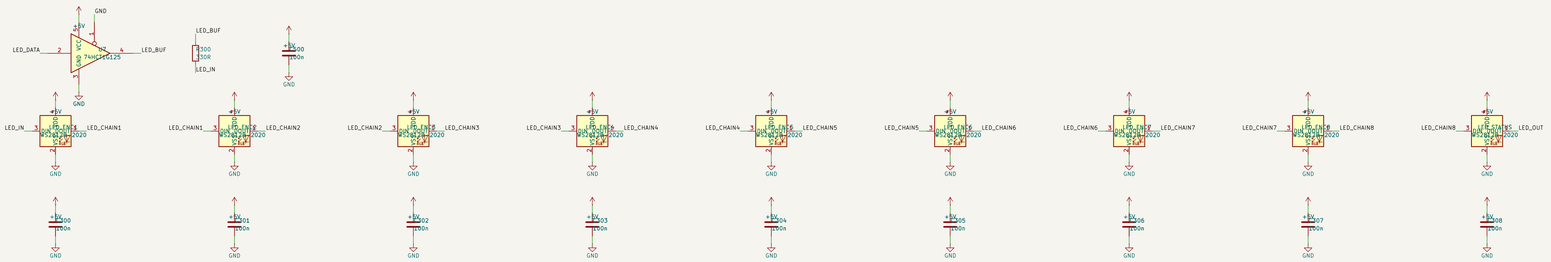
6 x 74HC165 cascade = 48 bits. CP=HC_CP(GP2) PL=HC_PL(GP6) QH=HC_QH(GP4/MISO). bits 0..15 encoders A/B, 16..23 encoder push, 24..27 shoulder, 28..47 reserved(GND).



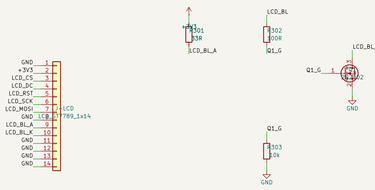
4 x shoulder button 6x6 mm -> 74HC165 bits 24..27 (BTN_L1/L2/R1/R2), active low, 10k pull-up



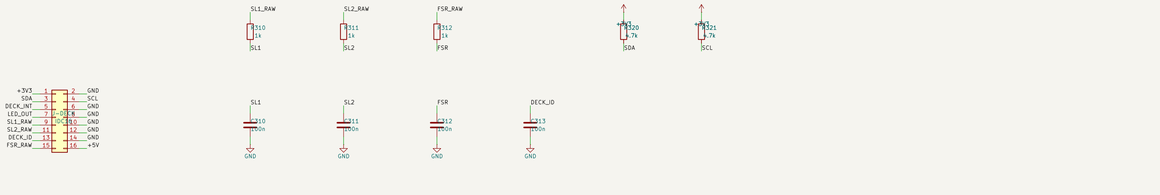
9 x WS2812B-2020 (8 knob rings + status) from +5V. GP7 (3V3) -> U7 74HCT1G125 (VCC = +5V, TTL input) -> R300 330R -> LED_IN ... LED_OUT -> deck chain. Chain current is limited by firmware brightness (spec/40). DECK_ID/DECK_INT: see notes at J-DECK/J-CORE.



LCD 2.8" ST7789 320x240 on SPI1 (SCK/MOSI), CS/DC/RST, BL via Q1 2N7002 (PWM GP15). Pin 1 is marked by the square pad; module pinouts differ — check the module before soldering the 1x14 header.



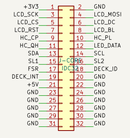
J-DECK IDC-16 pinout = spec/20-electronics.md 6.2 (odd pins = signal, even = GND). SL1/SL2 get the 1k/100n RC on PANEL-1 before the MCP3208 (CH0/CH1). FSR -> CH6, DECK_ID -> GP27, DECK_INT -> GP28 (spec/20 2: GP28 = DECK_INT, GATE1 = GP22, GATE2 = GP26).



J-CORE IDC-32 (2x16) -> CORE-1 J-PANEL: SPI1 display (SCK/MOSI/CS/DC/RST/BL), SPI0 SCK(HC_CP)+MISO(HC_QH), HC_PL, LED_DATA, I2C0 (SDA/SCL) + deck analog (SL1/SL2/FSR/DECK_ID/DE

PINOUT OF J-CORE (2x16, odd = signal / even = GND)

1	+3V3	2	GND
3	LCD_SCK	4	GND
5	LCD_CS	6	GND
7	LCD_RST	8	GND
9	HC_CP	10	GND
11	HC_QH	12	GND
13	SDA	14	GND
15	SL1	16	GND
17	FSR	18	GND
19	DECK_INT	20	GND
21	+5V	22	GND
23	GND	24	GND
25	GND	26	GND
27	GND	28	GND
29	GND	30	GND
31	GND	32	GND



Bukifon open hardware — CERN-OHL-S-2.0

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Title: Bukifon PANEL-1 — 300x132x1.6 — 8xEC11 / 6x74HC165 / ST7789 2

Size: A1 | Date:
Rev: 1.0
KiCad E.D.A. 10.0.4
Id: 1/1